

MATH70113 Simulation Methods for Finance

Assessed Coursework

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April 7, 2025

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1 Part 1: Barrier Option Greeks

In this section, we use the pathwise sensitivity method to estimate Delta and Vega for a down-and-out call option. The pathwise sensitivity method is only defined when the payoff function is continuous and piecewise differentiable. Therefore, before using this method, we apply the following linear estimation (shown in Figure 1 and 2).

Definition 1.

$$V \approx \mathbb{E}[e^{rT}(S_T - K)^+ f(\min_{t \in [0, T]} S_t)]$$

where

$$f(x) = \begin{cases} 0 & \text{if } x < B - \frac{\varepsilon}{2} \\ \frac{1}{\varepsilon}[x - (B - \frac{\varepsilon}{2})] & \text{if } B - \frac{\varepsilon}{2} \leq x \leq B + \frac{\varepsilon}{2} \\ 1 & \text{otherwise} \end{cases}$$

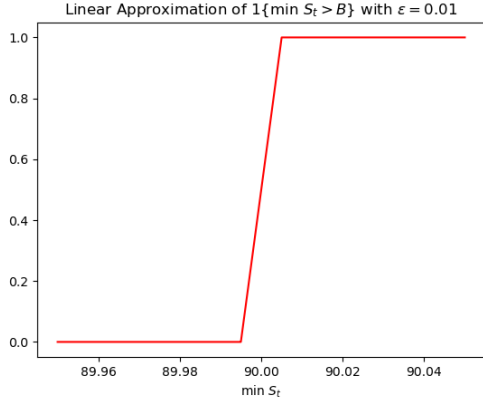


Figure 1: Linear Approximation of Indicator

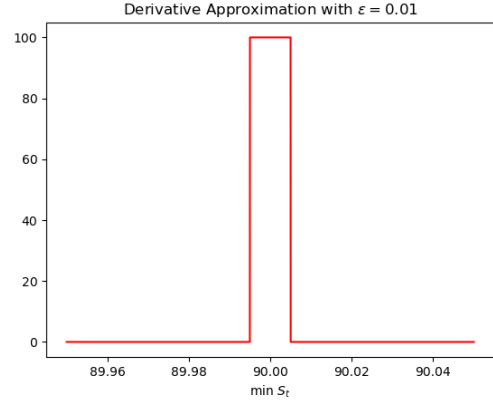


Figure 2: Approximation of Derivative

As a result, by interchanging the derivative and expectation, to calculate Delta and Vega we have

$$\begin{aligned} \Delta &= \frac{\partial V}{\partial S_0} \\ &\approx \mathbb{E}[e^{rT} \frac{\partial}{\partial S_0} ((S_T - K)^+ f(\min_{t \in [0, T]} S_t))] \\ &= \mathbb{E} \left[e^{rT} \left(1\{S_T > K\} \frac{\partial S_T}{\partial S_0} f(\min S_t) + (S_T - K)^+ f'(\min S_t) \frac{\partial \min S_t}{\partial S_0} \right) \right] \\ &= \mathbb{E} \left[e^{rT} \left(1\{S_T > K\} \frac{S_T}{S_0} f(\min S_t) + \frac{(S_T - K)^+}{\varepsilon} 1\left\{ \left| \min S_t - B \right| < \frac{\varepsilon}{2} \right\} \frac{\partial \min S_t}{\partial S_0} \right) \right] \\ \nu &= \frac{\partial V}{\partial \sigma} \\ &\approx \mathbb{E}[e^{rT} \frac{\partial}{\partial \sigma} ((S_T - K)^+ f(\min_{t \in [0, T]} S_t))] \\ &= \mathbb{E} \left[e^{rT} \left(1\{S_T > K\} \frac{\partial S_T}{\partial \sigma} f(\min S_t) + (S_T - K)^+ f'(\min S_t) \frac{\partial \min S_t}{\partial \sigma} \right) \right] \\ &= \mathbb{E} \left[e^{rT} \left(1\{S_T > K\} (S_T(-\sigma T + W_T)) f(\min S_t) + \frac{(S_T - K)^+}{\varepsilon} 1\left\{ \left| \min S_t - B \right| < \frac{\varepsilon}{2} \right\} \frac{\partial \min S_t}{\partial \sigma} \right) \right] \end{aligned}$$

where $S_t \triangleq S_0 \cdot e^{X_t}$ with $X_t = (r - \frac{\sigma^2}{2})t + \sigma W_t$.

These values are then estimated using Monte-Carlo. To allow us to simulate $\min_{t \in [0, T]} X_t$ we use inverse transform sampling, applied to the following conditional CDF:

Theorem 2. *Let (W_t) be a standard Brownian motion and define $Y_t = Y_0 + \mu t + \sigma W_t$. Then for $a \leq \min(Y_0, Y_T)$,*

$$P(\min_{t \in [0, T]} Y_t \leq a \mid Y_0, Y_T) = \exp\left(-\frac{2(Y_0 - a)(Y_T - a)}{\sigma^2 T}\right)$$

Hence, using the inverse transform method,

$$\min_{t \in [0, T]} X_t = \frac{X_t - \sqrt{X_t^2 - 2\sigma^2 T \ln(U)}}{2} \quad (1)$$

follows the desired distribution, where $U \sim \mathcal{U}(0, 1)$ and $\min_{t \in [0, T]} S_t = S_0 e^{\min_{t \in [0, T]} X_t}$

From this we have

$$\begin{aligned} \frac{\partial \min S_t}{\partial S_0} &= e^{\min_{t \in [0, T]} X_t} \\ \frac{\partial \min S_t}{\partial \sigma} &= \frac{-\sigma T + W_T}{2} - \frac{X_T(-\sigma T + W_T) - 2\sigma T \ln(U)}{2\sqrt{X_T^2 - 2\sigma^2 T \ln(U)}} \end{aligned}$$

We obtain Monte-Carlo estimates using $N = 10^8$ simulations and compare them to their analytical values. We note that it is not necessary to generate full discretized path simulations to generate these estimates. The values are shown below in Table 1.

Greek	Estimated value	Error
Delta	0.8500	0.076%
Vega	4.5221	-0.628%

Table 1: Estimated Greeks and Error for Down-and-Out Call Option (Pathwise Sensitivity)

2 Part 2: Review of “Multilevel Monte Carlo Path Simulation”

2.1 Introduction

Monte Carlo (MC) methods are widely used in finance to estimate values of stochastic processes and price a variety of exotic options. Yet the standard Monte Carlo approach can be computationally expensive, with a complexity of $O(\varepsilon^{-3})$ for a given target root-mean-squared-error (RMSE) ε . Michael B. Giles’ paper “Multilevel Monte Carlo Path Simulation” [5], introduces a new variance reduction technique, the Multilevel Monte Carlo method (MLMC). Using this easily implementable technique, computational complexity can be reduced to $O(\varepsilon^{-2} \log(\varepsilon)^2)$ at the expense of additional space requirements.

Giles’ findings extend the work of Kebaier [9], who showed computational cost can be reduced to $O(\varepsilon^{-2.5})$ and can be combined with other variance reduction techniques such as stratified sampling [7] and quasi-Monte Carlo methods [10] to obtain even greater savings.

In this report, we analyse Giles’ paper, providing an overview of the theoretical foundations of the MLMC method, before reproducing numerical results which demonstrate variance reduction when pricing a multitude of options. Lastly, we discuss related literature which explores other applications of the MLMC method.

2.2 Review of Paper

2.2.1 Theoretical Foundations

In computational finance, one is often interested in the expected value of a functional of the solution to a stochastic differential equation (SDE). Specifically, consider a multidimensional SDE with general drift and volatility terms and given initial data S_0 .

$$dS(t) = a(S, t) dt + b(S, t) dW(t), \quad 0 < t < T. \quad (2)$$

To compute the expected value of $f(S_T)$, where for example, $f(S)$ represents an option payoff function, the SDE is discretised

$$\hat{S}_{n+1} = \hat{S}_n + a(\hat{S}_n, t_n)h + b(\hat{S}_n, t_n)\Delta W_n,$$

and the simplest estimation for $E(f(S_T))$ is the mean of the payoff values $f(\hat{S}_{T/h})$, from N independent path simulations:

$$\hat{Y} = N^{-1} \sum_{i=1}^N f(\hat{S}_{T/h}^{(i)})$$

Provided $a(S, t)$ and $b(S, t)$ satisfy certain conditions, the expected mean-square error (MSE) in the estimate $\hat{Y}$ is of the asymptotic form

$$\begin{aligned} \text{MSE} &= E[(\hat{Y} - E[f(S_T)])^2] \\ &= E[(\hat{Y} - E[\hat{Y}])^2] + (E[\hat{Y}] - E[f(S_T)])^2 \\ &= E[(\hat{Y} - E[\hat{Y}])^2] + (E[f(\hat{S}_{T/h})] - E[f(S_T)])^2 \\ &\approx c_1 N^{-1} + c_2 h^2, \quad c_1, c_2 > 0. \end{aligned}$$

Therefore, making root-mean-square error (RSME) $O(\varepsilon)$ requires that $N = O(\varepsilon^{-2})$ and $h = O(\varepsilon)$, and hence the computational complexity (cost) is $O(\varepsilon^{-3})$.

2.2.2 Multilevel Monte Carlo

Consider Monte Carlo path simulations with different timesteps $h_l = M^{-1}T, l = 0, \dots, L$ for integer $M \geq 2$ and $L \geq 1$. Given a Brownian path $W(t)$, we denote the payoff $f(S_T)$ by P , and let $\hat{S}_{l,M^l}$ and $\hat{P}_l$ denote the approximations to $S(T)$ and P using a numerical discretisation with timestep h_l . Through the telescoping sum we get

$$E[\hat{P}_L] = E[\hat{P}_0] + \sum_{l=1}^L E[\hat{P}_l - \hat{P}_{l-1}].$$

The MLMC method estimates each expectation on the above equation's right-hand side independently, in a way that minimises computational complexity.

We define $\hat{Y}_0$ be the simple MC estimator for $E[\hat{P}_0]$ using N_0 samples, and $\hat{Y}_l$ for $l > 0$ to be the simple MC estimator for $E[\hat{P}_l - \hat{P}_{l-1}]$ using N_l paths. Specifically

$$\hat{Y}_l = N_l^{-1} \sum_{i=1}^{N_l} (\hat{P}_l^{(i)} - \hat{P}_{l-1}^{(i)}). \quad (3)$$

It is important to emphasize here that the quantity $\hat{P}_l^{(i)} - \hat{P}_{l-1}^{(i)}$ is computed using the same underlying Brownian path but using different timesteps. Going forward, we will refer to these as the fine and coarse discretisations respectively. The variance of this simple estimator is $V[\hat{Y}_l] = N_l^{-1}V_l$, where V_l is the variance of an individual sample. Hence, the combined estimator $\hat{Y} = \sum_{l=0}^L \hat{Y}_l$ has variance

$$V[\hat{Y}] = \sum_{l=0}^L N_l^{-1}V_l,$$

which gives a computational cost proportional to $\sum_{l=0}^L N_l h_l^{-1}$. For a fixed cost, variance is minimised by choosing N_l proportional to $\sqrt{V_l h_l}$, which could be proved using Lagrange multipliers.

For a Lipschitz payoff function, provided $a(S, t)$ and $b(S, t)$ satisfy certain conditions [3], the Euler discretisation has $O(h)$ weak convergence and $O(h^{1/2})$ strong convergence. Therefore, as $l \rightarrow \infty$,

$$E[\hat{P}_l - P] = O(h_l) \quad (4)$$

and

$$E[\|\hat{S}_{l,M^l} - S(T)\|^2] = O(h_l). \quad (5)$$

Since $a(S, t)$ and $b(S, t)$ are Lipschitz, it follows that

$$V[\hat{P}_l - P] \leq E[(\hat{P}_l - P)^2] \leq c^2 E[\|\hat{S}_{l,M^l} - S(T)\|^2],$$

and so $V[\hat{P}_l - P] = O(h_l)$. Furthermore, as

$$V[\hat{P}_l - \hat{P}_{l-1}] \leq ((V[\hat{P}_l - P])^{1/2} + (V[\hat{P}_{l-1} - P])^{1/2})^2,$$

it follows that the single sample variance $V(l)$ is $O(h)$ in the simple estimator (3) and the optimal choice for N_l is asymptotically proportional to h_l . Therefore, if we set $N_l = O(\varepsilon^{-2} L h_l)$ the variance of the combined estimator $\hat{Y}$ is $O(\varepsilon^2)$. L is chosen so that

$$L = \frac{\log \varepsilon^{-1}}{\log M} + O(1).$$

As $\varepsilon \rightarrow 0$, $h_L = M^{-L} = O(\varepsilon)$, and due to equation (4), the bias error $E[\hat{P}_L - P]$ is $O(\varepsilon)$. Hence, by using the standard bias-variance decomposition, the mean-square error (MSE) is $O(\varepsilon^2)$, with a computational complexity $O(\varepsilon^{-2} L^2) = O(\varepsilon^{-2} (\log \varepsilon)^2)$.

2.2.3 Complexity Theorem

The Complexity Theorem is the central theorem of Giles' paper, and it quantifies the computational advantage of the MLMC method over the standard MC method. It is stated generally, making it applicable to a variety of financial derivatives where payoff functions may be discontinuous or path-dependent, for example, digital and lookback options. The proof, is provided in Giles' paper [5].

Theorem 3 (Complexity Theorem). *Let P denote a functional of the solution of SDE (2) for a given Brownian path $W(t)$, and $\hat{P}_l$ denote the corresponding approximation using numerical discretisation with timestep $h_l = M^{-l}T$.*

If there exist independent estimators $\hat{Y}_l$ based on N_l Monte Carlo samples, and positive constants $\alpha > \frac{1}{2}, \beta, c_1, c_2, c_3$ such that

- (i) $\mathbb{E}[|\hat{P}_l - P|] \leq c_1 h_l^\alpha$ (weak order α),
- (ii) $\mathbb{E}[\hat{Y}_l] = \begin{cases} \mathbb{E}[\hat{P}_0], & l = 0, \\ \mathbb{E}[\hat{P}_l - \hat{P}_{l-1}], & l > 0, \end{cases}$
- (iii) $\mathbb{V}[\hat{Y}_l] \leq c_2 N_l^{-1} h_l^\beta$,
- (iv) C_l , the computational complexity of $\hat{Y}_l$, is bounded by $C_l \leq c_3 N_l h_l^{-1}$,

then there exists a positive constant c_4 such that for any $\varepsilon < e^{-1}$, there are values L and N_l for which the multilevel estimator

$$\hat{Y} = \sum_{l=0}^L \hat{Y}_l$$

has an MSE with bound

$$MSE = \mathbb{E}[(\hat{Y} - \mathbb{E}[P])^2] < \varepsilon^2,$$

with a computational complexity C with bound

$$C \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > 1, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = 1, \\ c_4 \varepsilon^{-2-(1-\beta)/\alpha}, & 0 < \beta < 1. \end{cases}$$

The theorem demonstrates the importance of the coefficient β , which determines the convergence of $\mathbb{V}(\hat{Y}_l)$ as $l \rightarrow \infty$. The optimal N_l is proportional to $\sqrt{V_l h_l} = O(h_l^{(\beta+1)/2})$, and so the computational complexity is proportional to $O(h_l^{(\beta-1)/2})$. Therefore, the computational cost is mostly expended on the coarsest levels when $\beta > 1$, the finest levels when $\beta < 1$, and approximately uniformly when $\beta = 1$.

For a Lipschitz payoff, the Euler discretisation achieves $O(h)$ weak convergence and $O(h^{1/2})$ strong convergence, and hence $\beta = 1$ as explained earlier. The convergence deteriorates when the payoff function $f(S(T))$ has a discontinuity. In such cases, for timestep h_l , the probability that S_{l,M^l} and $S_{l-1,M^{l-1}}$ lie on different sides of the discontinuity is $O(h_l^{1/2})$. Hence $\hat{P}_l - \hat{P}_{l-1}$ is $O(1)$ with probability $O(h_l^{1/2})$ and $\mathbb{V}(\hat{Y}_l)$ is $O(h_l^{1/2})$ instead of $O(h_l)$. This means that the computational complexity increases from $O(\varepsilon^{-2}(\log \varepsilon)^2)$ to $O(\varepsilon^{-2.5})$, but it is still an improvement from the original $O(\varepsilon^3)$.

2.3 Numerical Results

2.3.1 Numerical Algorithm

To reproduce the results presented in the paper, we implement the provided multilevel algorithm, which collates the aforementioned theoretical results for practical use as follows:

- Step 1.* Start with $L = 0$.
- Step 2.* Estimate V_L using an initial set of $N_L = 10^4$ samples.
- Step 3.* Define optimal N_l , $l = 0, \dots, L$, using equation (6).
- Step 4.* Evaluate extra samples at each level as needed for new N_l .
- Step 5.* If $L \geq 2$, test for convergence using equation (7).
- Step 6.* If $L < 2$ or it is not converged, set $L := L + 1$ and go to Step 2.

The equation used for optimal N_l is

$$N_l = \left\lceil 2\varepsilon^{-2} \sqrt{V_l h_l} \left(\sum_{l=0}^L \sqrt{V_l / h_l} \right) \right\rceil. \quad (6)$$

The equation used to determine convergence is

$$\max \left\{ M^{-1} |\hat{Y}_{L-1}|, |\hat{Y}_L| \right\} < \frac{1}{\sqrt{2}} (M - 1) \varepsilon. \quad (7)$$

Here, we see optimal N_l is proportional to $\sqrt{V_l h_l}$ and equation (6) ensures the combined multilevel estimator $\hat{Y}$ has an estimated variance of less than $\frac{1}{2}\varepsilon^{-2}$, given that the estimate V_l is accurate enough. Equation (7) attempts to keep bias less than $\frac{1}{\sqrt{2}}\varepsilon$. Together, they should yield a MSE less than ε^2 , where ε is a user-specified root mean squared accuracy, yet it is important to emphasize that the algorithm is heuristic and not guaranteed to achieve an $O(\varepsilon^2)$ MSE. The Complexity Theorem does guarantee this, however its conditions rely on a priori knowledge of the constants c_1 and c_2 , which govern the weak convergence and variance convergence as $h \rightarrow \infty$. These constants are implicitly estimated within the numerical algorithm described above.

We also consider Richardson extrapolation, which attempts to correct the terminal bias by adding the term $(M - 1)^{-1} \hat{Y}_L \approx (M - 1)^{-1} E[\hat{P}_L - \hat{P}_{L-1}] \approx E[P - \hat{P}_L]$. The estimator becomes

$$\frac{M}{M - 1} \left\{ \hat{Y}_0 + \sum_{l=1}^L (\hat{Y}_l - M^{-1} \hat{Y}_{l-1}) \right\},$$

with the termination condition

$$| \hat{Y}_l - M^{-1} \hat{Y}_{l-1} | < \frac{1}{\sqrt{2}} (M^2 - 1) \varepsilon. \quad (8)$$

Using Euler discretisation and simple geometric Brownian motion,

$$dS(t) = rS(t)dt + \sigma S(t)dW(t), \quad 0 < t < 1,$$

with $S(0) = 1, r = 0.05, K = 1$ and $\sigma = 0.2$, we now present results using the MLMC numerical algorithm for 4 different options. For every value of ε , we also estimate the RMSE by repeating the estimation process 10 times, and present the range of values of RMSE/ε obtained for each option.

2.3.2 European Call

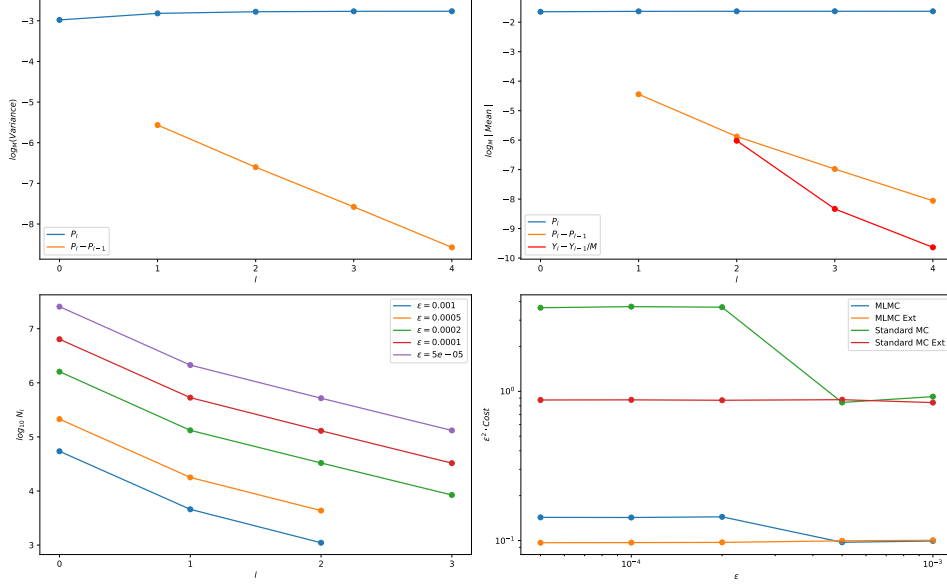


Figure 3: Plots for European call option under geometric Brownian dynamics.

Figure 3 shows results for the European call option with discounted payoff function

$$P = e^{-r} \max(S(1) - 1, 0)$$

The two graphs in the first row are generated with 4×10^6 paths for all levels, independent of the actual MLMC algorithm and are used to illustrate convergence properties. The top-left graph illustrates the behaviour of the variance for both $\hat{P}_l$ and $\hat{P}_l - \hat{P}_{l-1}$, plotted logarithm base M (where $M = 4$, as in the paper). This gives a slope close to -1 , indicating that $V_l = \mathbb{V}[P_l - P_{l-1}] = O(h)$. The top-right plot illustrates the mean value and correction at each level, again the slope close to -1 indicates $O(h)$ convergence of $E[\hat{P}_l - \hat{P}_{l-1}]$.

The bottom-left plot illustrates N_l values for varying values of ε and is decreasing in l due to the decrease in both V_l and h_l . We also observe that as ε decreases, the maximum level L increases.

The bottom-right plot illustrates how computational cost C varies for different values of ε , for both the MC and MLMC methods. We define computational cost for the MLMC method as:

$$C = N_0 + \sum_{l=1}^L N_l (M^l + M^{l-1}),$$

and compare this to the computational cost for the standard MC method, defined as:

$$C^* = \sum_{l=0}^L N_l^* M^l,$$

where $N_l^* = 2\varepsilon^{-2}\mathbb{V}[P_l]$. In this case, we assume that Monte Carlo is performed on each level with the same stopping condition as before. This choice of N_l ensures that when the stopping condition is satisfied, the variance of the standard Monte Carlo estimator is $\frac{1}{2}\varepsilon^2$ as in the multi-level method. This plot demonstrates the computational superiority of the MLMC method and empirically verifies the theoretical results stated in the Complexity Theorem. For a European call option, the theory states that $C = O(\varepsilon^{-2}(\log \varepsilon)^2)$, which is consistent with our observation that $\varepsilon^2 C$ is only very weakly dependent on ε . For $\varepsilon = 10^{-4}$ the MLMC method is ten times more efficient than the standard MC method, and this difference increases for smaller ε .

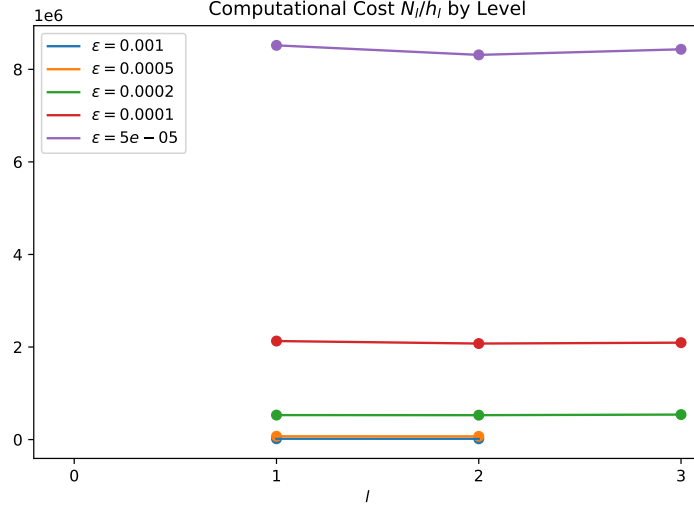


Figure 4: Computational cost for European call for different levels

Since $\beta = 1$, the computation cost $O(h_l^{(\beta-1)/2})$ is expected to be expended uniformly across all levels, as verified in Figure 4. The RMSE/ε ratio is smaller than 1 across all values of ε with and without the Richardson extrapolation, suggesting that the MLMC method consistently achieves the given accuracy requirement.

2.3.3 Fixed Strike Asian Call

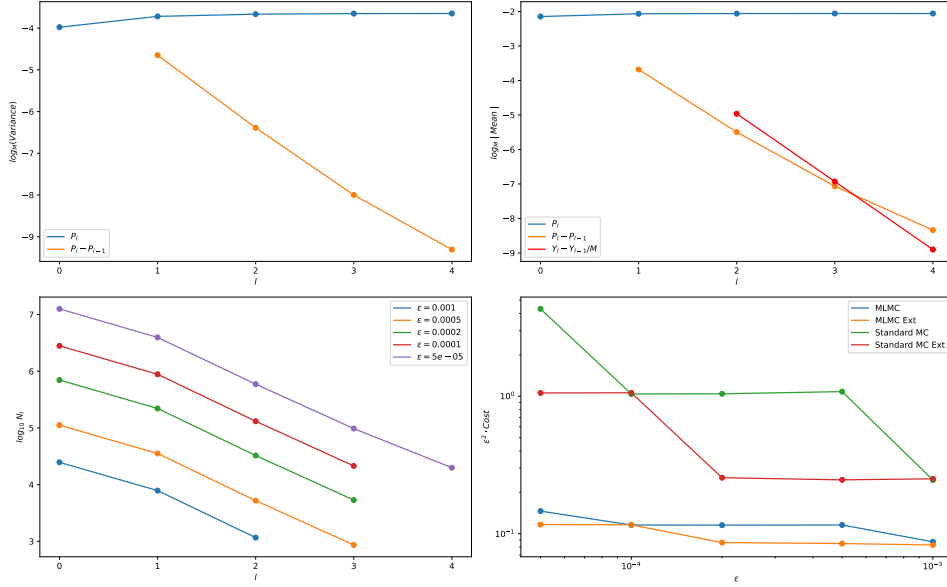


Figure 5: Plots for Asian call option under geometric Brownian dynamics.

Figure 5 shows results for the Asian option with payoff $P = e^{-r} \max(\bar{S} - 1, 0)$, where

$$\bar{S} = \int_0^1 S(t) dt.$$

For a given l , we numerically approximate this integral by

$$\bar{S}_l = \sum_{n=1}^{N_l} \frac{1}{2} (\hat{S}_n + \hat{S}_{n-1}) h_l.$$

Again, for both V_l and $E[P_l - P_{l-1}]$, we observe $O(h_l)$ convergence. However, we observe that the RMSE/ϵ deteriorates in this case, and is decreasing with respect to ϵ . This ratio even reaches 7 for the smallest ϵ level, which suggests that the Asian option prices obtained using MLMC are much less reliable.

2.3.4 Floating Strike Lookback Call

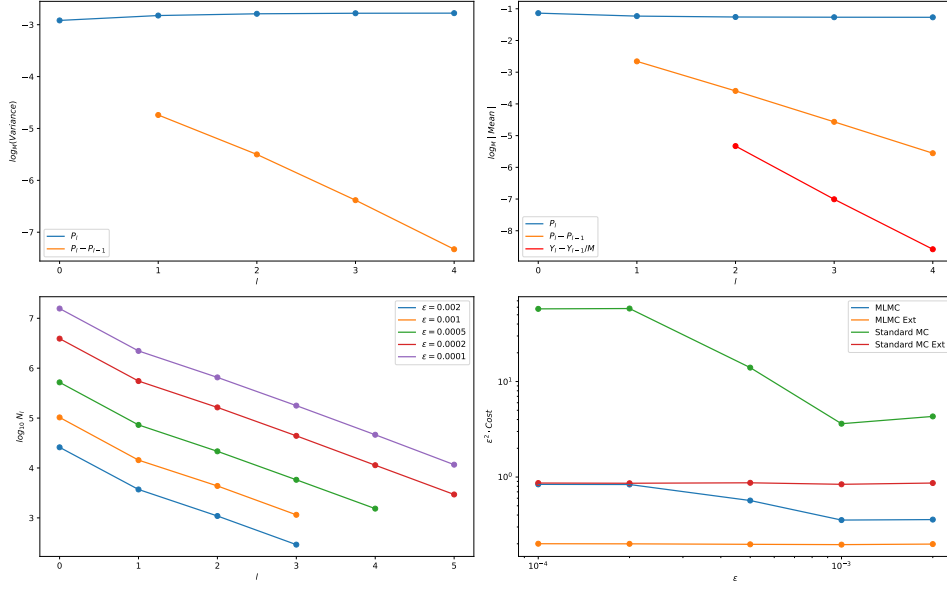


Figure 6: Plots for floating strike lookback call under geometric Brownian dynamics.

Figure 6 shows results for the floating strike lookback call option with payoff

$$P = e^{-r}(S_1 - \min_{0 \leq t \leq 1} S_t)$$

where we approximate the running minimum by

$$\min_{0 \leq t \leq 1} S_t \approx (\min_n \hat{S}_n)(1 - \beta^* \sigma \sqrt{h_l}) \quad (9)$$

where $\beta^* = 0.5826$ is constant. The factor in the RHS of this equation corrects for the discretization error when computing the minimum value on a given sample path, thus restoring $O(h)$ convergence, which is evident in the plots [4]. While this option is also path-dependent, the empirical RMSE is much better than the Asian options, with only one violation with a RMSE/ϵ ratio of 1.32 at the smallest level of ϵ .

2.3.5 Digital Call

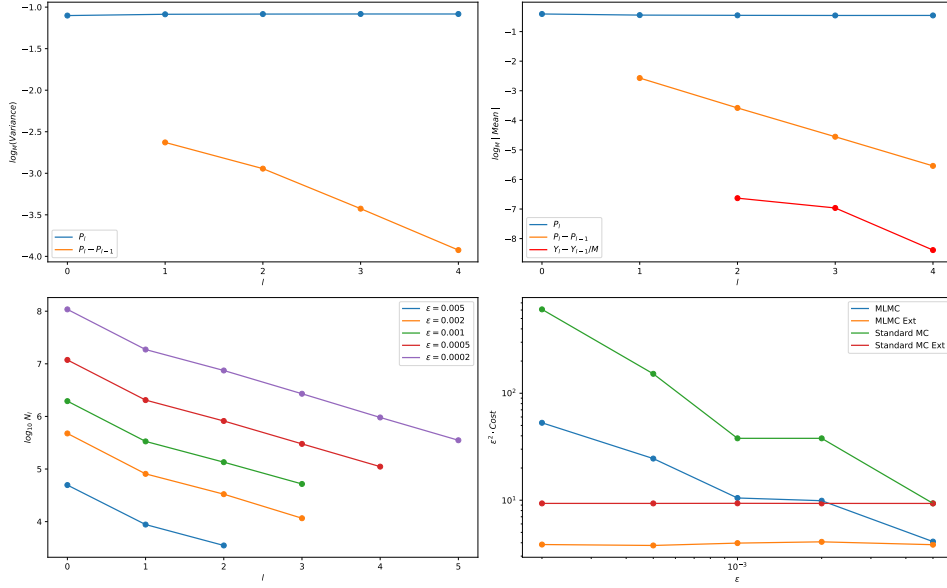


Figure 7: Plots for digital call option under geometric Brownian dynamics.

Figure 7 shows results for the digital call option with payoff $P = e^{-r} \mathbf{1}_{\{S(1) \geq 1\}}$.

Given the discontinuous payoff of the digital option, $V_l = O(h_l^{1/2})$, displayed by the slope of approximately -0.5 in the top-left plot. As a result, considerably larger values of N_l are required to achieve comparable accuracy and the effectiveness of the MLMC method is reduced.

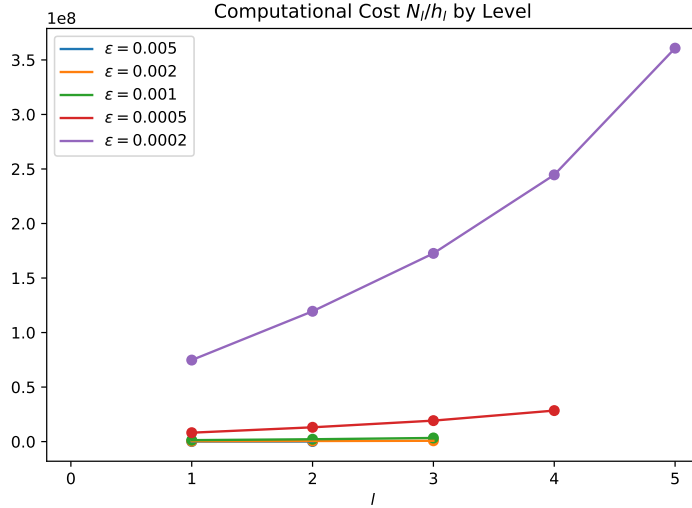


Figure 8: Computational cost for digital call for different levels.

Since $\beta < 1$, the computational cost $O(h_l^{(\beta-1)/2})$ is decreasing in h_l , and hence is focused on the finer grids (Figure 8). RMSE/ϵ takes values in the range $0.33 - 1.15$, achieving the desired accuracy for larger values of ϵ in most cases. We note that for all options considered, RMSE/ϵ violations are much more frequent for small values of ϵ .

2.3.6 Heston Stochastic Volatility Model

We now present results for the European call option using the Heston stochastic volatility model:

$$dS = rS dt + \sqrt{V}S dW_1, \quad 0 < t < 1,$$

$$dV = \lambda(\sigma^2 - V) dt + \xi\sqrt{V} dW_2,$$

with $S(0) = 1$, $V(0) = 0.04$, $r = 0.05$, $\sigma = 0.2$, $\lambda = 5$, $\xi = 0.25$, and correlation $\rho = -0.5$ between dW_1 and dW_2 . For better accuracy and variance when implementing, we introduce a new variable $W = e^{\lambda t}(V - \sigma^2)$ and apply Euler discretisation to the SDEs:

$$\hat{S}_{n+1} = \hat{S}_n + r\hat{S}_nh + \sqrt{\hat{V}_n^+}\hat{S}_n\Delta W_{1,n},$$

$$\hat{V}_{n+1} = \sigma^2 + e^{-\lambda h} \left((\hat{V}_n - \sigma^2) + \xi\sqrt{\hat{V}_n^+}\Delta W_{2,n} \right).$$

To ensure stability, we replace $\sqrt{V}$ with $\sqrt{V^+} \equiv \sqrt{\max(V, 0)}$. However, with our choice of parameters λ , σ , and ξ , the probability of the discretised volatility process becoming negative tends to zero, as $h \rightarrow 0$. [8]

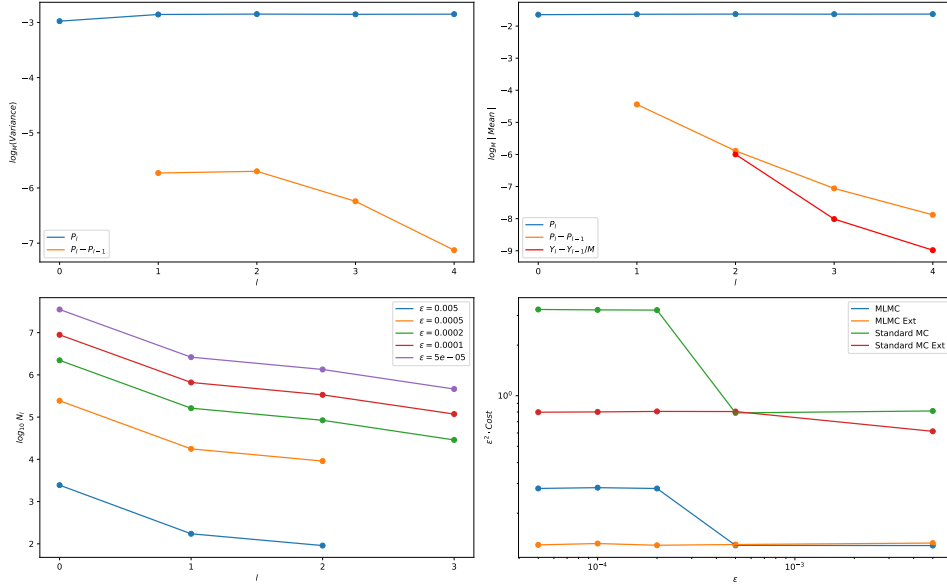


Figure 9: Plots for European call option under Heston stochastic volatility dynamics.

Here, the Complexity Theorem is not directly applicable as volatility does not satisfy a global Lipschitz condition. Nonetheless, the numerical results indicate slightly slower than first order variance decay, but a marginally faster than first order weak convergence. The MLMC's significant computational savings compared to the standard MC method are also evident.

Although the MLMC method offers a noticeable reduction in computational cost, its memory requirement is significantly larger in practice. For standard Monte Carlo, it is sufficient to keep track of the running sum of the realisations of the target random variable. However, the MLMC method requires storing all the values of $\hat{P}_l^{(i)} - \hat{P}_{l-1}^{(i)}$ in order to recompute V_l at every level and could be a challenge in practice. This recomputation might also prevent the improvement in computational complexity from being reflected in the algorithm runtime.

2.4 Extensions

In this section we explore related literature on additional applications of the MLMC method. We begin by discretising the stock process using a Milstein scheme in order to study the effects this has on strong convergence for different options. Following this, we examine the MLMC method within the context of quantum computing.

Specifically, we explore Giles' subsequent paper "Improved Multilevel Monte Carlo Convergence using the Milstein Scheme" [6] before moving onto a related paper title "Quantum-accelerated multilevel Monte Carlo methods for stochastic differential equations in mathematical finance" [1].

2.4.1 Milstein Discretisation

The Milstein discretisation of equation (2) is

$$\hat{S}_{n+1} = \hat{S}_n + a(\hat{S}_n, t_n) h + b(\hat{S}_n, t_n) \Delta W_n + \frac{1}{2} \frac{\partial b}{\partial S} b(\hat{S}_n, t_n) [(\Delta W_n)^2 - h].$$

Milstein discretisation achieves $O(h)$ strong convergence, leading to faster variance decay for options with Lipschitz payoffs. It also brings additional benefits for path-dependent or discontinuous options, which we explore in this section's analysis.

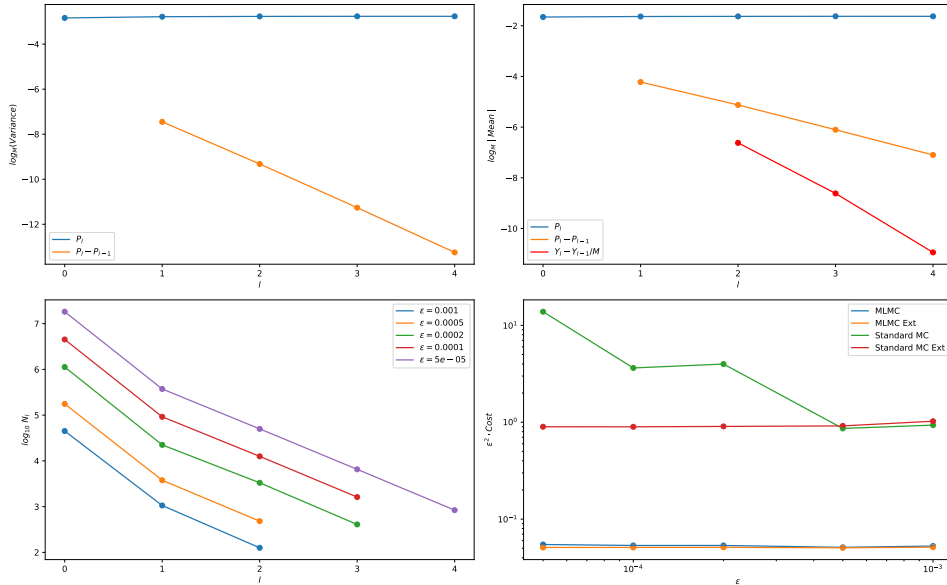


Figure 10: Plots for call option under geometric Brownian dynamics with Milstein discretisation.

When applied to a European call option, the decrease in variance is obvious, with the slope of -2 representing the improvement of V_l from $O(h_l)$ to $O(h_l^2)$. The top right plot is relatively unchanged, a consequence of the identical weak convergence of the Euler and Milstein scheme. Since $\beta = 2$, the number of paths per level decreases much quicker with respect to l compared to the Euler scheme, and the computational cost is concentrated on the coarse levels. The theoretical $O(\epsilon^2)$ computation cost shown in the Complexity Theorem is also verified in the bottom-right plot, with $\epsilon^2 \cdot \text{cost}$ being independent of the specified RMSE. We note that although the number of levels has increased for the smallest value of ϵ , the overall cost has decreased. The MLMC method is much more efficient compared to the regular Monte Carlo method, with

the difference in computational cost being greater than a factor of 100, which is larger than the difference obtained with Euler discretisation.

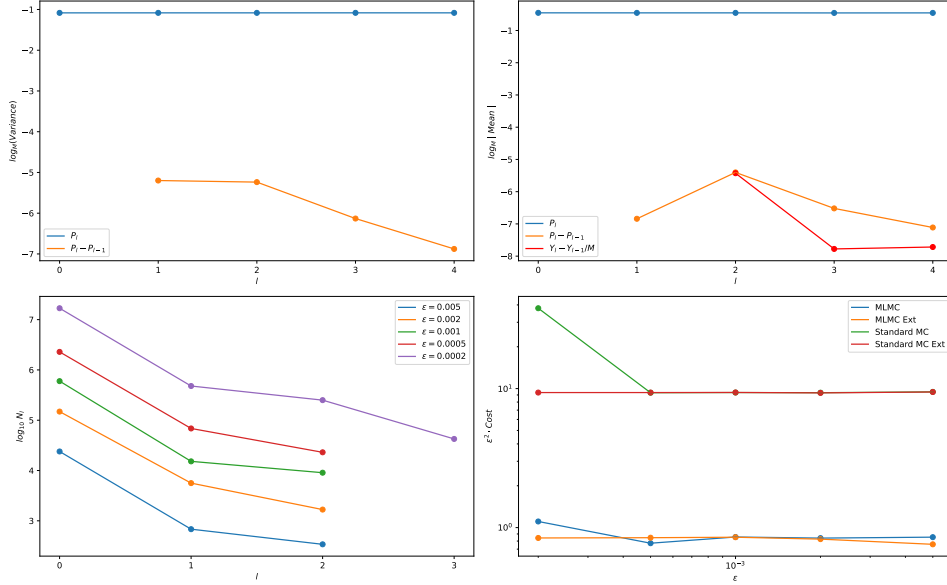


Figure 11: Plots for digital call option under geometric Brownian dynamics with Milstein discretisation.

For the digital call, while the increased order of strong convergence is not reflected in the top-right plot due to the discontinuity, the mean and variance of $P_l - P_{l-1}$ is lower than before for all values of l . This allowed the algorithm to terminate much earlier and with a much lower computation cost.

However, due to the improved strong convergence, the order of magnitude of $|\hat{S}_{l,M^l} - \hat{S}_{l-1,M^{l-1}}|$ is much smaller, which combined with the piecewise-constant payoff function, could cause our estimate of V_l to be 0. To prevent this from occurring, it is necessary to perform an additional check before using Equation 6. In this case, if V_l is zero for any l , we set it to the smallest non-zero variance estimator in that iteration. While the Milstein discretisation converges much quicker, the additional cost of computing the Milstein adjustment term in practice is non-negligible and is not reflected in the computational cost C .

2.4.2 Quantum Computing

With the main advantage of Multilevel Monte Carlo over classical Monte Carlo being a reduced computational complexity, it is natural to study how the complexity can be further reduced whilst still retaining the same order of accuracy in the estimated value. A candidate algorithm that achieves this is Quantum Accelerated Multilevel Monte Carlo (QA-MLMC) [2], combining the MLMC algorithm structure with the speed-up achieved by Quantum Amplitude Estimation (QAE) when performing Monte Carlo estimates of the quantities $\mathbb{E}[\hat{P}_l - \hat{P}_{l-1}]$.

Though out of the scope of this report, the main idea of the Quantum Monte Carlo algorithm is to encode the sample payoffs simulated with the numerical scheme, into the probability amplitudes of each state in a qubit register. This, in turn, allows the use of QAE to ‘encode’ the quantity we want to estimate into the eigenfrequency of the oscillating quantum register,

which we can then retrieve with an ‘ancilla’ qubit system. A more detailed explanation of this algorithm can be found on [11]. The remarkable result is that this entire process achieves a complexity of $\mathcal{O}(\varepsilon^{-1})$, as opposed to the classical Monte Carlo complexity of $\mathcal{O}(\varepsilon^{-2})$ [11].

Using the results from Theorem 3 and the speed-up achieved by QAE, in [2] the following Theorem is proven.

Theorem 4 (QA-MLMC for Option Pricing). *Consider the problem of wanting to estimate $\mathbb{E}[\mathcal{P}(S_T)|S_t = S_0]$ for a final piecewise Lipschitz continuous payoff $\mathcal{P}$ and a suitable quantum oracle for Quantum Amplitude Estimation. Quantum Accelerated Multilevel Monte Carlo with a numerical scheme for SDEs of strong order r estimates $\mathbb{E}[\mathcal{P}(S_T)|S_t = S_0]$ up to additive error ε with probability at least 0.99 with complexity*

$$\begin{cases} \mathcal{O}(\varepsilon^{-1}), & r > 2, \\ \mathcal{O}(\varepsilon^{-1/2-1/r-o(1)}), & r \leq 2. \end{cases}$$

Additionally, if the payoff function $\mathcal{P}$ is globally Lipschitz continuous everywhere, then the complexity can be improved to

$$\begin{cases} \mathcal{O}(\varepsilon^{-1}), & r \geq 1, \\ \mathcal{O}(\varepsilon^{-1/r}), & r < 1. \end{cases}$$

Though the Euler-Maruyama numerical scheme is only of strong convergence $r = \frac{1}{2}$, the previously discussed Milstein scheme does satisfy $r = 1$. Hence, QA-MLMC under the Milstein scheme is a drastic quadratic improvement from MLMC for pricing options with Lipschitz continuous everywhere payoffs.

To implement the QA-MLMC algorithm, we make use of the QISKIT Python library, allowing us to run the QAE algorithm for the necessary Monte Carlo computation steps. It is worth noting, however, that due to the dimensions of the Hilbert space spanned by the superposition of states of the qubit register, the values N_l must now be of the form $N_l = 2^k$ for some $k \in \mathbb{N}$ representing the number of qubits. As a result of this, we modify Equation (6) into

$$N_l^{Quantum} = 2^{\lceil \log_2(2\varepsilon^{-2}\sqrt{V_l h_l}(\sum_{l=0}^L \sqrt{V_l/h_l})) \rceil}, \quad (10)$$

to ensure that the simulation can be encoded into the qubit register. Nevertheless, the large numbers of $N_l^{Quantum}$ for each iteration, implying simulating a growing large number of qubits, make the algorithm extremely slow on a classical computer for any satisfactory value of ε , hence, no meaningful results can be extracted from this section.

2.5 Conclusion

In our replication of the Multilevel Monte Carlo method presented by Giles [5], we find that the algorithm is effective in reducing computational complexity when used with European, Asian, Digital and Lookback options. Indeed, there are significant computational savings compared to a standard Monte Carlo approach. However, the MLMC has a notable practical limitations. Substantial memory is required to perform and store simulations across multiple refinement levels, and so broader applications of this methodology require extensive computational resources making the technique relatively less accessible.

In extending the MLMC method using a Milstein discretisation, we find further improvements in the convergence rate of the MLMC algorithm. However, the computation of the Milstein adjustment terms are not factored in to this convergence rate, once again introducing additional computational costs. Our attempts at utilizing the quantum amplitude estimation algorithm to improve the convergence of the MLMC estimator were less fruitful. While in theory, the method should enable meaningful improvement in the convergence rate we find that the procedure is too computationally intensive to simulate without additional resources.

We believe future areas of research can include incorporating quasi-Monte Carlo methods to reduce computational complexity, and extending the algorithm to multi-dimensional SDEs. Furthermore, it would be valuable to test the MLMC method on the considerably more complex payoff structures of some of the products that currently exist in financial markets (e.g. best-of and worst-of options on baskets), to broaden its robustness and utility.

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